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Art and Machine Learning

20 April 2026

Visualization of Sound and its Motion

ABSTRACT

Music and visual art both convey emotion. This study acts as a bridge between the two to map music to visual art in a consistent and systematic way. The audio to visual pipeline extracts low level audio features (RMS, spectral centroid, ZCR, bass/treble) and identifies the mood of the musical input using a small classifier trained on the DEAM dataset from a wide variety of music. These features are mapped to visual elements (circles, stars, 3-7 sided polygons, spirals, bursts, and flowers) and rendered frame by frame in a MP4. This creates a visual synchronized with the audio. This pipeline has been able to create visuals for a wide range of music, including classical music such as Bach's Suite No. 3, pop music such as Bruno Mars' Locked Out of Heaven, and electronic music such as Skrillex's Redline Dash. To make sure the audience can understand what emotions and movement the visualizer is trying to represent, there is also a mood graph next to the visualizer that presents the changes over time for both the mood of the music and the low level audio features.

INTRODUCTION

Sound-based art, such as music, and visual art, such as film, are both powerful mediums through which artists express emotion, depict meaningful scenes, and communicate complex ideas. Although they are often experienced separately, both forms aim to create deep perceptual and emotional connections with an audience. People naturally form associations between sounds and visual elements. For example, higher-pitched sounds like a flute or birdsong are often linked to lightheartedness, while deeper sounds, such as that from an organ, tend to evoke darker and heavier imagery. These patterns suggest that humans intuitively develop cross-sensory connections, revealing an underlying relationship between auditory and visual perception, with visual perception often taking the dominant role.

The trend of visuals dominating the foreground and audio being a supporting element is prevalent across many forms of media. Taking the Star Wars film series as an example, the main things audiences tend to focus on are the character interactions, action sequences, and the visual spectacle, with the music and sound effects serving as a way to further enhance the mood and immersiveness of each scene. This is also seen in interactive media such as open world video games, where the music accompanies the player as they explore, dynamically changing based on the player location and action.

This study aims to invert this relationship, making sound the element that drives the narrative while the visuals serve as a response and accompaniment. This allows audiences to experience the structure and movement of sound in a visual space, offering a new way to interpret the ideas, emotions, and stories that musical pieces capture.

CONCEPT

The central question of this project was: how can we visualize sound in a way that conveys the space it takes up, consistently produces meaningful results, and communicates emotion from raw audio data? Addressing this took an iterative process, progressing from rudimentary feature-mapping experiments toward a system capable of reflecting the full emotional and structural complexity of music.

Early stages began with mapping audio into a visual form based on features such as volume, pitch, and frequency onto a dynamic graph, making it possible to trace variations across a song and identify which elements carried the least or most amount of change. This led to two visualizers: a morphing flower where color and shape responded to volume, a central circle reflected bass presence, and petal length corresponded to higher pitches, and a grid-based system where shapes and colors shifted with volume, note, and tone.

The final visualizer takes form as two parts, the first producing a generative image visualizer and the second being an emotional hexagon visualizer. The two are in sync and complement one another in understanding the music.

The generative image visualizer places various shapes (stars, spirals, squares, and polygons) of varying color, opacity, and size at various points on the canvas on the beat of the music, collectively producing a collage like plotted artwork that reflects the variance and sonic character of a given piece. Similar to plotting a graph, volume determines x-position and size, pitch determines y-position and color, note duration is reflected through opacity, and audio texture is represented by shape type. Audio analysis and feature extraction were conducted using the Python library Librosa, which incorporates machine learning methods for signal processing.

The extracted audio values were subsequently passed through algorithms and randomizers, developed with input from AI tools including ChatGPT and Microsoft Copilot, to ensure that equivalent audio changes do not always produce identical visual responses. This results in controlled variation into the output, reducing overlap and increasing the diversity of what the visualizer produces.

The second component shifts focus toward the emotional dimension of music. Each song is segmented into 0.2 second intervals, with each interval evaluated across six audio dimensions (energy, pitch, bass, treble, valence, and arousal) represented on a dynamic hexagonal graph. Emotion labels are assigned based on a valence arousal mapping framework: high valence combined with high arousal corresponds to ‘excited’; high valence with low arousal to ‘calm’; low valence with high arousal to ‘tense’; and low valence with low arousal to ‘sad’. This classification is based on the established valence arousal model of emotional representation.

To systemize this, a Convolutional Neural Network (CNN) was trained on the DEAM (MediaEval Database for Emotional Analysis in Music) dataset to output valence and arousal directly from data of audio input. The pipeline follows the structure: audio file → mel spectrogram → CNN → valence and arousal calculation → emotional label output. This architecture facilitates the hexagonal radar plot to shift dynamically and in sync throughout the track, generating a cohesive emotional portrait of the song as it progresses. To keep the radar synchronized with playback without flickering or stalling, predicted values are precomputed into a 10 Hz sample table and indexed by video time using a forward walking cursor.

Side by side, the two components support and address the original question. Part 1 captures the characteristics of structure through generative visual art, while part 2 captures the emotional trajectory through machine learning driven mood classifications.

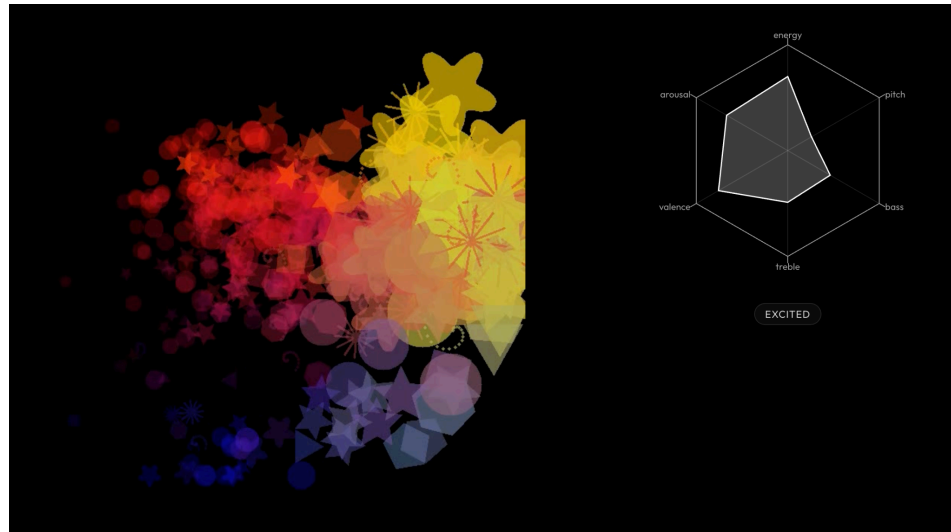


Figure 1. Generative Image Visualizer and Emotional Hexagon Visualizer

DATASET

We used the Database for Emotional Analysis of Music (DEAM), a publically annotated corpus developed by Aljanki et al. (2017), to train a classifier capable of assigning emotional labels to music. DEAM contains 1,802 music excerpts spanning a wide range of genres, including classical, jazz, pop, rock, and electronic. They were each rated by multiple crowdsourced workers on two continuous dimensions: valence (how positive or negative the music feels) and arousal (how energetic or calm it feels). Specifically, both dimensions were recorded on a nine-point Likert scale with annotations averaged over two second intervals.

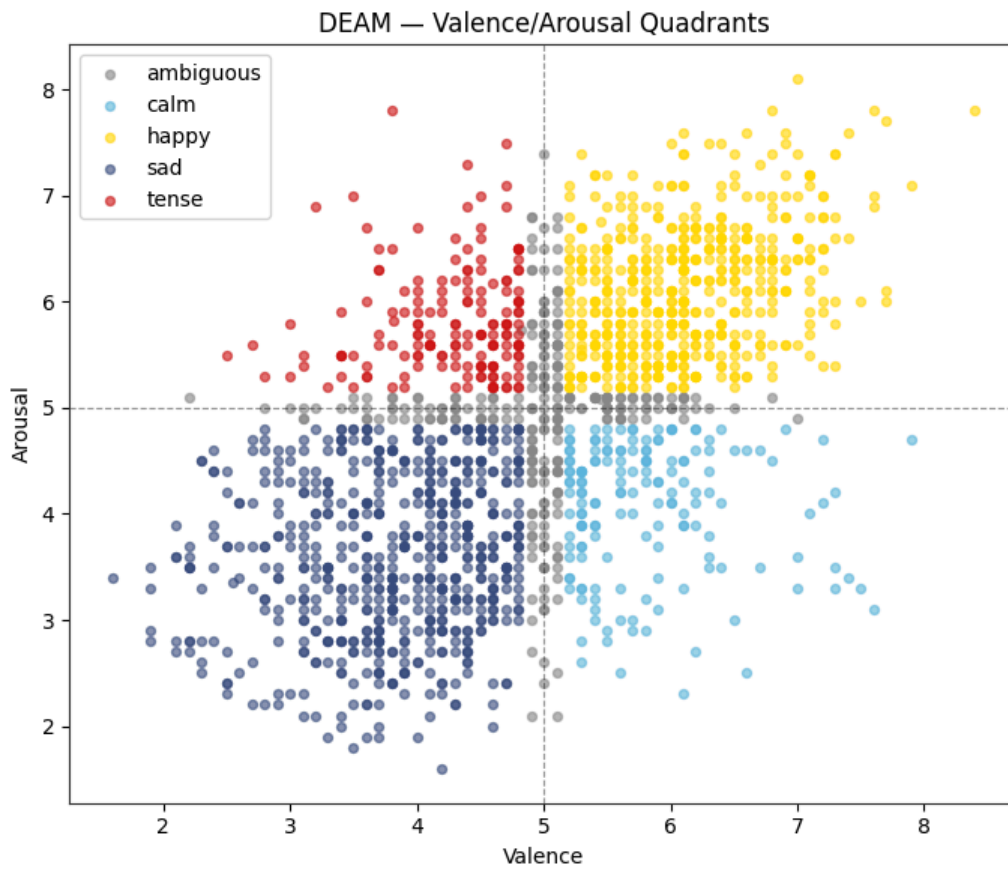
The downloaded archive contains three components: raw MP3 audio files at 44,100 Hz (MEMD_audio), pre-extracted hand-crafted features (features), and annotation CSVs at both temporal resolutions (DEAM_Annotations). Note that we only used the audio and song-level annotations since the CNN model described in the next section learned its own representation directly from the mel-spectrogram input.

Table 1. Summary of the DEAM dataset used for emotional classifier training

Property	Value
Total Annotated Songs	1,744
Clip Duration	45 seconds each
Annotators Scale	1-9 Likert (valence, arousal)
Annotators Per Song	~10 (Amazon Mechanical Turk)
Audio Format	MP3, 44,100 Hz mono
Genres	Classical, jazz, pop, rock, electronic, etc.

Since DEAM’s annotations are continuous rather than categorical, we proceeded with a labelling step to convert each song’s mean valence and arousal scores into one of four discrete emotion classes: happy, calm, tense, and sad. Following Russell’s Circumplex Model of Affect (Russell, 1980), the valence-arousal plane is divided into four quadrants at the scale midpoint of 5.0. Furthermore, songs within ± 0.2 units of the midpoint on either axis are withheld as ambiguous and excluded from training to avoid noise.

Figure 2. Mapping of valence-arousal quadrants to emotion labels



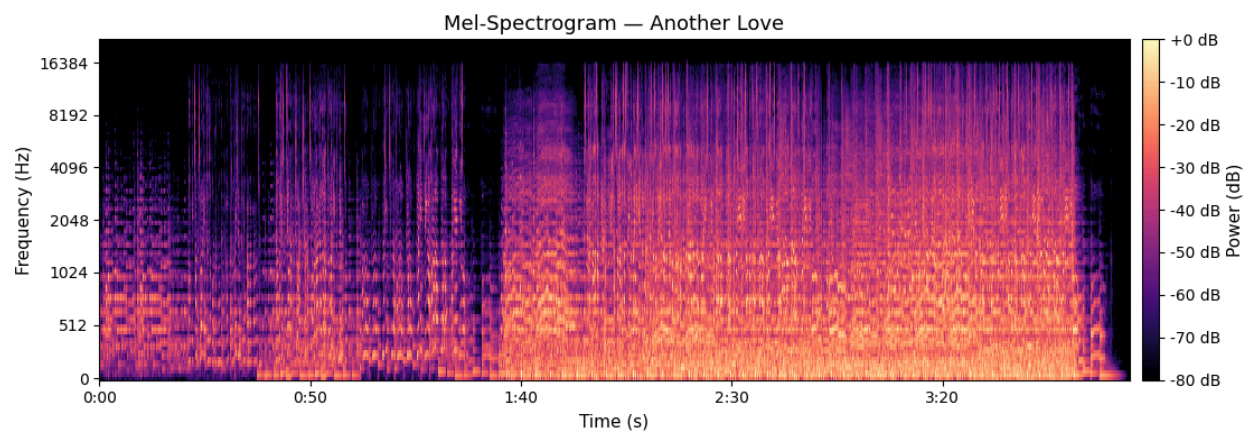
The margin of 0.2 was chosen empirically after inspecting the distribution of songs across several reasonable values. At a margin of 0.2, approximately 85 percent of songs received a label, with 623 sad, 526 happy, 272 ambiguous, 164 tense, and 159 sad. On the other hand, a margin of 0.5 reduced the number of calm and tense examples to fewer than 15% of the corpus, which caused the classifier to collapse onto happy and sad predictions.

MODEL

For our inputs, rather than feeding raw waveforms or hand-crafted features into the classifier, we instead converted each audio segment into a mel-spectrogram, which is a perceptual scale that approximates the human auditory system’s non-linear frequency resolution. We decided to choose this because mel-spectrograms have been well established as effective CNN inputs for audio classification tasks (Pons et al., 2017) because they make harmonic structure, tempo, and timbral texture simultaneously visible as spatial patterns.

Each song is divided into non-overlapping five-second windows. Then every window is converted into a mel-spectrogram with 128 mel bins, an FFT window of 2,048 samples, and a hop length of 512 samples. Finally, the spectrogram is converted to decibels using the `power_to_db` method from Librosa.

Figure 3. Mel-Spectrogram of “Another Love” by Tom Odell



With our inputs ready, we built a Convolutional Neural Network that treats each mel-spectrogram segment as a greyscale image and regresses two continuous outputs: normalized valence and normalized arousal, both scale to the range $[0, 1]$ by a `MinMaxScaler`

fitted on the training split. The sigmoid activation on the output layer enforces this range. The Mean Squared Error is used as the training loss, with Mean Absolute Error reported as the primary interpretable metric.

The network consists of four convolutional blocks followed by two fully connected layers. Each convolutional block applies a 3x3 convolution with same-padding and ReLU activations, followed by Batch Normalization and 2x2 Max Pooling. After the fourth block, a Global Average Pooling collapses the spatial dimensions into a 256-dimensional feature vector, which is passed through two dense layers with dropout regularization before the two-unit output head.

Table 2. CNN architecture

Layer	Configuration	Output Shape	Parameters
Conv2D block 1	32 filters, 3x3, same, ReLU + BN + Pool 2x2	6 x 215 x 32	448
Conv2D block 2	64 filters, 3x3, same, ReLU + BN + Pool 2x2	32 x 107 x 64	18,752
Conv2D block 3	128 filters, 3x3, same, ReLU + BN + Pool 2x2	16 x 53 x 128	74,368
Conv2D block 4 + GAP	256 filters, 3x3, same, ReLU + BN + GAP	256	295,936
Dense 1 + Dropout	256 units, ReLU, dropout $p = 0.4$	256	65,792
Dense 2 + Dropout	128 units, ReLU, dropout $p = 0.3$	128	32,896
Output	2 units, sigmoid	2	258

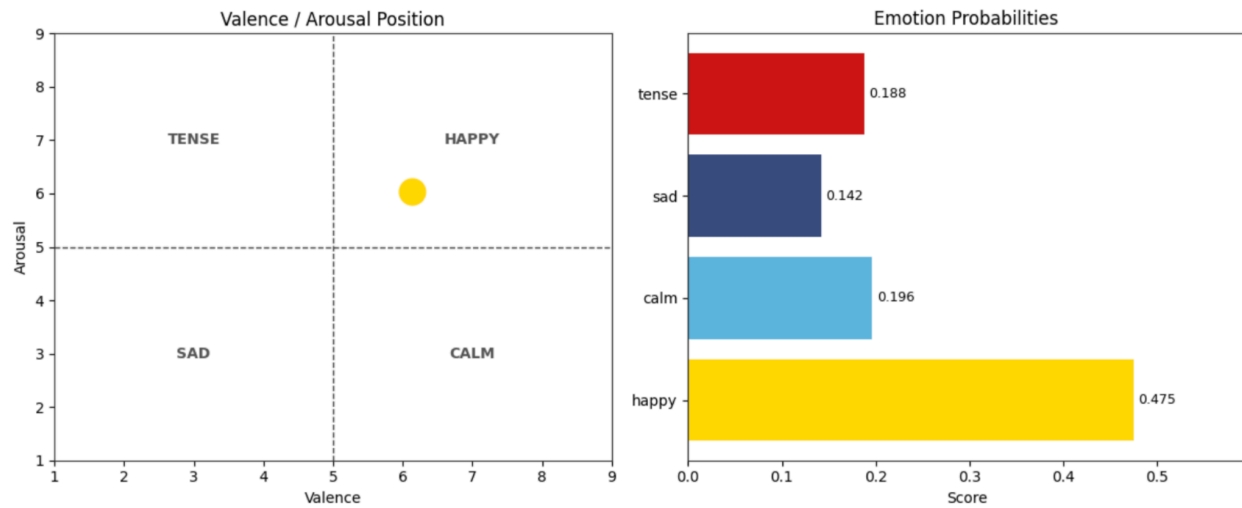
Total trainable parameters: approximately 490,000.

The model is then trained using the Adam optimizer with an initial learning rate of 0.001 on the oversampled, balanced training set. We setup three callbacks to regulate training: (1) early stopping with patience of ten epochs that restores the best validation weights, (2) model checkpointing saving only improvements in validation MSE, and (3) ReduceLROnPlateau halving the learning rate after five epochs without improvement, down to a minimum of 10^{-6} . A maximum of sixty epochs is chosen, though early stopping typically triggers between epochs 35 and 50.

At inference time, the full audio file is segmented into five-second windows using the same procedure as training. The CNN predicts a normalized (valence, arousal) pair for each window. These predictions are then averaged across all windows, and the mean is inverse-transformed by the saved MinMaxScaler to recover the values on the original 1-9 scale. The resulting point is then passed to the same quadrant assignment function used during labelling, yielding a single emotional label for the entire track.

To provide additional interpretability, we included a confidence score using Euclidean distance from the predicted point (valence, arousal) to the midpoint (5,5), normalized to the range [0,1], with higher values indicating more unambiguous predications farther from the decision boundary. Also, we added soft probability scores for each of the four quadrants using inverse Euclidean distances from the predictions to each quadrant centroid: (7,7) for happy, (7,3) for calm, (3, 3) for sad, and (3,7) for tense.

Figure 4. Predicted Valence-Arousal Position and Emotion Probabilities for “No Broke Boys” by Disco Lines, Tinashe



RESULTS

Link to video website: <https://artml.vercel.app/>

The following images are results of the Generative Image Visualizer.



[Pitbull, Don't Stop The Party]



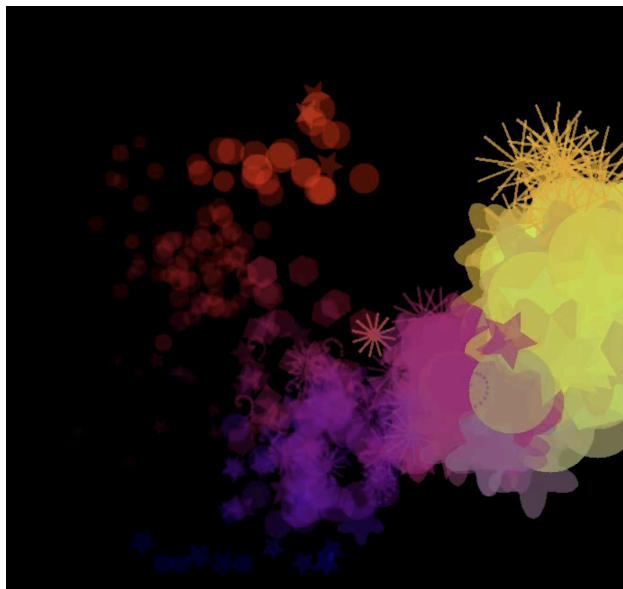
[Zombie Nation, Kernkraft 400]



[BIGBANG, IF YOU]



[Tony Britten, UEFA Champions League Anthem]

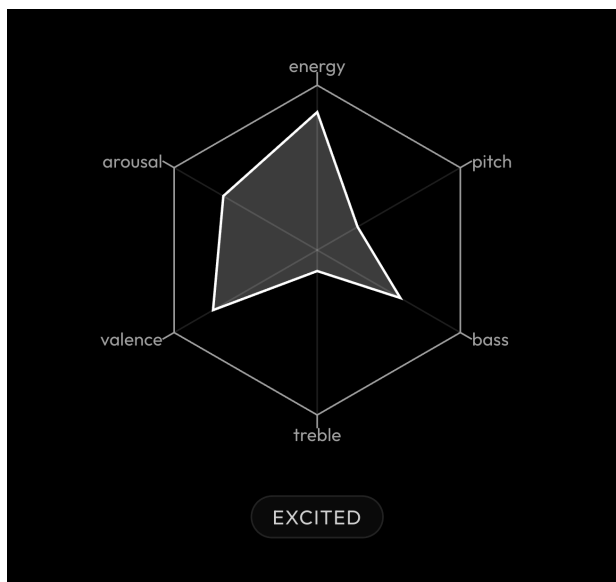


[Skrillex, REDLINE DASH]

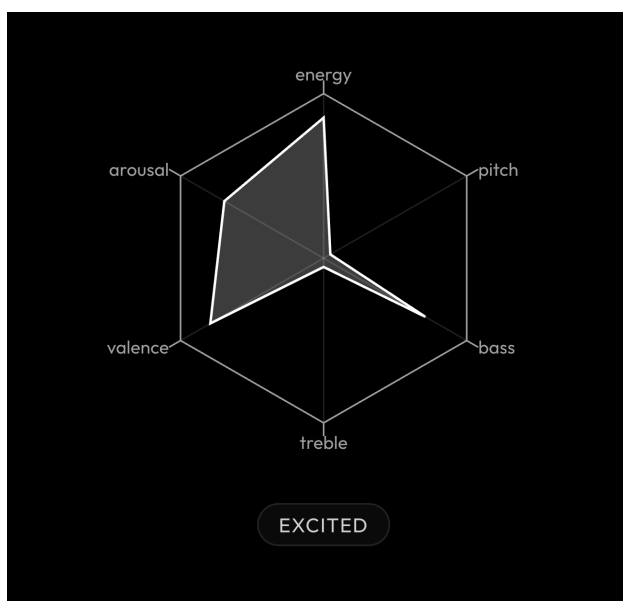


[J.S. Bach, Suite No. 3]

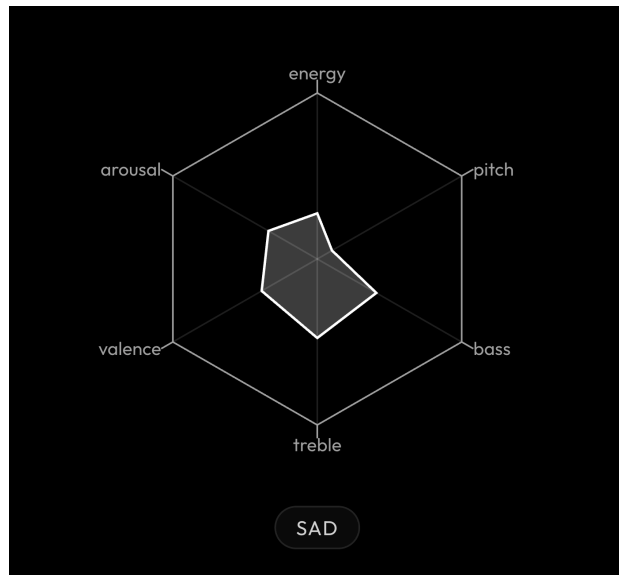
The following images are results of the Emotional Hexagon Visualizer.



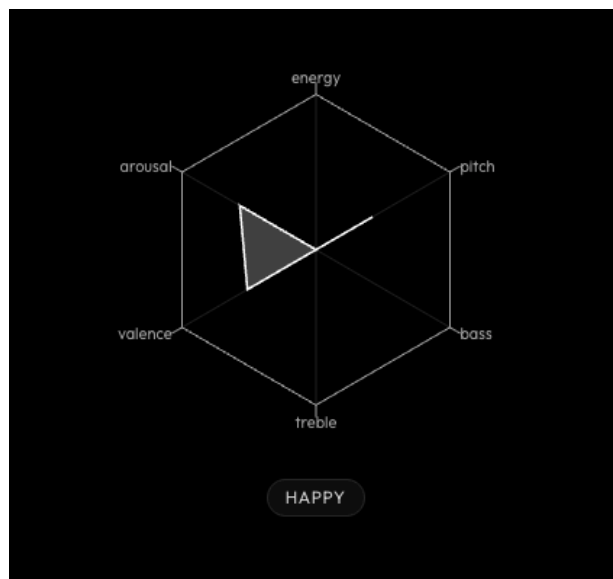
[Pitbull, Don't Stop The Party]



[Skrillex, REDLINE DASH]



[J.S. Bach, Suite No. 3]



[BIGBANG, IF YOU]

To test the output of the visualizer, we generated videos showing the progression of the visualizer over time for orchestral, pop, and electronic music. Presented above are the last frames of the video for each song, from which a few trends can be noticed. For Bach's Suite No. 3 and Britten's UEFA Champions League Anthem, the shapes are mostly red and orange that are

largely concentrated in one region near the middle of the graph with the occasional yellow outlier representing occasional crescendos or peaks in the music. This indicates a relatively stable sound texture with few sudden changes in the energy, pitch, and other audio features, as is common with ceremonial music and slow lyrical movements (such as the “Air” movement from the Baroque Era).

In contrast, Pitbull’s Don’t Stop the Party and Zombie Nation’s Kernkraft 400 covers more space and has a larger range of colors, representing the various types of sounds and pitches in the piece. Deeper sounds of the electronic music along with human vocals creates a diverse landscape that contrasts those of classical music. Although Skrillex’s REDLINE DASH is also electronic music, it creates a different cluster compared to the two. The presence of quick and high pitched notes creates smaller shapes as seen through the small red shapes. The short, high pitched, and frivolous sounds, when doubled with deeper tones, produce a fluctuating and ecstatic visualizer characterized by an element of unpredictability that evolves into a repetitive rhythmic pattern. All three were prominently labeled energetic throughout the song, but produced varying final visualizers

Lastly, BIGBANG’s IF YOU is also represented by shapes that are more spread out across the canvas, but compared to electronic sounds, like those of Pitbull and Skrillex, this song has a higher concentration of oranges and red shapes than blue and purple, similarly to the classical pieces sampled. The use of instruments throughout the piece as to deep electronic sounds could be attributed to the top heavier concentration. Despite its sad lyrics and emotional qualities, the higher pitched acoustic sounds paired with faster paced rhythmic stability categorizes it as happy, proving that perceived emotion and acoustic features don’t always align.

Classical music is characterized by melodic, fluid, and connected sounds, in comparison, electronic music is often focused on sharper contrasts and repetitive beats with deeper notes. IF YOU acts as a middle ground between classical music and electronic music with its use of instruments and repetitive quicker segmented sounds that comes together to create a right heavy but also top concentrated final visualizer.

FUTURE CONSIDERATIONS

Future work can expand the system in multiple ways. One direction could be testing the pipeline on AI-generated music from tools like Suno or Udio, where songs are prompted with specific emotional intent. Using our CNN model can help test whether the AI created music from Suno or Udio actually conveys the emotion it was prompted with. A different extension of the system can involve incorporating key detection into the emotion pipeline. This can help improve mood deception. On the presentation side, the demo presented to the class included precomputed videos; it would be interesting to see video selection at runtime - the user can select any song they like and the pipeline would generate the video in real time. This would help improve interactivity with the art. Another direction to take would be having user-tunable visual mappings; letting users adjust how features map to visuals (e.g., choosing their own color palette for valence, or switching shape mappings).

CONCLUSION

Overall, this visualizer was able to effectively translate musical features such as the pitch, treble/bass, energy level, and overall positivity/negativity of the music into visual form. However, since the mood was only determined using the arousal and valence as threshold values,

there is a limit on the numbers of emotional descriptors we could use to characterize the mood of the music. In addition, we were not able to capture more complex features of the music such as the key, the musical mode (including major/minor), and the instrumentation, which are often important factors in determining the emotions within music. Future versions of this visualizer system can benefit from incorporating more of these features as data, which can help in creating a more nuanced visual translation that can better represent the space and movement of the music.

Works Cited

- Aljanaki, A., Yang, Y.-H., & Soleymani, M. (2017). Developing a benchmark for emotional analysis of music. *PLOS ONE*, 12(3), e0173392.
- Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161–1178.
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